

WEEK 3 ASSIGNMENT

ENTITY FRAMEWORK CORE 8.0

LAB-1

1. What is ORM?

- ORM maps C# classes to database tables
Benefits: Productivity, maintainability, abstraction from SQL

2. EF Core vs EF Framework:

- EF Core is cross-platform, lightweight, and supports modern features like LINQ, async queries, and compiled queries
EF Framework (EF6) is Windows-only, more mature but less flexible

3. EF Core 8.0 Features:

- JSON column mapping
- Improved performance with compiled models
- Interceptors and better bulk operations

4. Create a .NET Console App:

- `dotnet new console -n RetailInventory`
- `cd RetailInventory`

5. Install EF Core Packages:

- `dotnet add package Microsoft.EntityFrameworkCore.SqlServer`
- `dotnet add package Microsoft.Entity.p`

LAB-2

C# Category.cs X

Models > C# Category.cs

```
1  public class Category {  
2      public int Id { get; set; }  
3      public string Name { get; set; }  
4      public List<Product> Products { get; set; }  
5  }  
6
```

C# Category.cs

C# Products.cs X

Models > C# Products.cs

```
1 public class Product {  
2     public int Id { get; set; }  
3     public string Name { get; set; }  
4     public decimal Price { get; set; }  
5     public int CategoryId { get; set; }  
6     public Category Category { get; set; }  
7 }  
8
```

Category.cs X

AppDbContext.cs X

AppDbContext.cs

```
1 public class AppDbContext : DbContext {
2     public DbSet<Product> Products { get; set; }
3     public DbSet<Category> Categories { get; set; }
4
5     protected override void OnConfiguring(DbContextOptionsBuilder optionsBuilder) {
6         | optionsBuilder.UseSqlServer("Your_Connection_String_Here");
7     }
8 }
9
```

OUTPUT

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter

```
[Running] scriptcs "c:\Users\KIIT\RetailInventory\AppDbContext.cs"  
'scriptcs' is not recognized as an internal or external command,  
operable program or batch file.
```

```
[Done] exited with code=1 in 0.032 seconds
```

LAB-3

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\KIIT\RetailInventory> dotnet build
>> dotnet ef migrations add InitialCreate
>> dotnet ef database update
>>
Determining projects to restore...
All projects are up-to-date for restore.
RetailInventory -> C:\Users\KIIT\RetailInventory\bin\Debug\net8.0\
RetailInventory.dll

Build succeeded.
    0 Warning(s)
    0 Error(s)

Time Elapsed 00:00:00.80
Build started...
Build succeeded.
```

LAB-4

Program.cs X

Program.cs

```
1  using var context = new AppDbContext();
2
3  var electronics = new Category { Name = "Electronics" };
4  var groceries = new Category { Name = "Groceries" };
5
6  await context.Categories.AddRangeAsync(electronics, groceries);
7
8  var product1 = new Product { Name = "Laptop", Price = 75000, Category = electronics };
9  var product2 = new Product { Name = "Rice Bag", Price = 1200, Category = groceries };
10
11  await context.Products.AddRangeAsync(product1, product2);
12  await context.SaveChangesAsync();
13  |
```

LAB-5

Program.cs X

Program.cs

```
1  using var context = new AppDbContext();
2
3  // 1. Retrieve all products
4  var products = await context.Products.ToListAsync();
5  foreach (var p in products)
6  |      Console.WriteLine($"{p.Name} - ₹{p.Price}");
7  |
8  // 2. Find by ID
9  var product = await context.Products.FindAsync(1);
10 Console.WriteLine($"Found: {product?.Name}");
11
12 // 3. FirstOrDefault (e.g., expensive product)
13 var expensive = await context.Products.FirstOrDefaultAsync(p => p.Price > 50000);
14 Console.WriteLine($"Expensive: {expensive?.Name}");
15 |
```

OUTPUT

PROBLEMS

OUTPUT CONSOLE

TERMINAL

PORTS

○ PS C:Users\KIIT\RetailInventory> **dotnet** run

Laptop - ₹75000

Rice Bag - ₹1200

Found: Laptop

Expensive: Laptop

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